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USE OF THE KIDNEY FAILURE RISK EQUATION TO INFORM CLINICAL CARE OF PATIENTS WITH CHRONIC KIDNEY DISEASE: A MIXED-METHODS SYSTEMATIC REVIEW

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USE OF THE KIDNEY FAILURE RISK EQUATION TO INFORM CLINICAL CARE OF PATIENTS WITH CHRONIC KIDNEY DISEASE: A MIXED-METHODS SYSTEMATIC REVIEW

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ABSTRACT

Rationale & Objective: The Kidney Failure Risk Equation (KFRE) predicts the risk of end-stage renal disease in patients with chronic kidney disease (CKD). This study aimed to evaluate the evidence base for the utility of KFRE in clinical practice.

Study Design: Systematic review.

Setting & Study Populations: Adult patients with CKD but not receiving renal replacement therapy enrolled in studies where KFRE was used in clinical care pathways.

Selection Criteria for Studies: All studies published from April 2011 to February 2021 identified from Medline, CINAHL, Embase and reference and citation searches of included studies.

Data Extraction: Relevant data were extracted, and two reviewers independently assessed study quality using appropriate appraisal tools.

Analytical Approach: Findings reported as a narrative synthesis due to heterogeneity of the included studies.

Results: Of 1425 studies identified, 367 duplicates were removed. One thousand and fifty-eight titles and abstracts were screened. Five studies underwent a full-text review, and all were included in the analysis. Three uses of KFRE were assessed in these studies: i) at the

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primary to speciality care interface; ii) general nephrology to multidisciplinary care transition; iii) treatment planning.

Limitations: The quality of included studies was variable. Limited data in two studies made it difficult to assess if the population was representative and findings generalisable.

Conclusions: This is the first systematic review of studies that have assessed the actual impact of KFRE in clinical practice. To date, five studies of varying quality have been reported, evaluating the use of KFRE in clinical care pathways. Trials are in progress assessing the impact on clinical outcomes of using KFRE in clinical practice, and KFRE is being incorporated into guidelines for CKD management. Further studies are needed to assess the impact of KFRE on clinical care.

Registration: Protocol registered on PROSPERO before initiation of the study (Ref: CRD42020219926).

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- To the best of our knowledge, this is the first systematic review to examine the evidence for the impact of the use of the Kidney Failure Risk Equation (KFRE) in clinical practice.
- A mixed-methods approach was used to allow the inclusion of both quantitative and qualitative evidence.
- Study quality ranged widely, and some studies did not provide adequate detail of their population characteristics, such that generalisability was difficult to assess.
- Further, the statistical analysis was limited in several studies.

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BACKGROUND

The global prevalence of chronic kidney disease (CKD) is estimated to be 9.1% and CKD was the 12th leading cause of death in 2017.[1] CKD is also associated with an increased risk of progression to end-stage renal disease (ESRD). Risk stratification in CKD can enable more efficient care, with specialty care targeted to patients at higher risk of ESRD, whilst sparing those with low risk unnecessary intervention and undue anxiety associated with this.[2,3]

Most guideline criteria for referral of patients to specialist nephrology care use estimated glomerular filtration rate (eGFR), eGFR decline, and urine albumin:creatinine ratio (ACR) thresholds rather than a quantified ESRD risk.[4] Such criteria often result in the referral of patients at low risk of ESRD and the non-referral of patients at high risk.[5] Therefore, recent CKD guidelines endorse risk-based thresholds for specialty referral and renal replacement therapy (RRT) planning.[6,7]

The best validated risk prediction model is the Kidney Failure Risk Equation (KFRE), which predicts the 2- and 5-year risk of ESRD in patients with stages 3-5 CKD and has undergone extensive validation.[8-10] The equation can easily be embedded into electronic medical records and is also readily available online.[11]

It is uncertain whether the use of the KFRE in clinical practice has a meaningful impact on clinical pathways and health outcomes and how patients and healthcare professionals view the KFRE. We carried out a systematic review of the available evidence of the impact of the use of the KFRE in clinical practice.

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METHODS

The study protocol was registered on PROSPERO (Ref: CRD42020219926), and the study is reported as per the 'Preferred Reporting Items for Systematic Review and Meta-Analysis' (PRISMA) checklist.[12]

Data Sources and Searches:

Two reviewers (HKB and AF) independently searched MEDLINE (Ovid), Cumulative Index to Nursing and Allied Health Literature (CINAHL) (EBSCO) and Embase (Ovid) for studies between April 2011 to February 2021 that examined the impact of utilising the KFRE in patients with CKD.

The search strategies (Table S1) were developed with the support of an information specialist and used keywords, index terms and MeSH terms tailored and applied to each individual database. No language restrictions were applied. We also hand searched the reference lists of included studies and performed a forward citation review of these studies and the original KFRE development study[13]. All results obtained at each stage of the process were entered into EndNote version X9.3.3 (Clarivate Analytics, PA, USA), and duplicates were removed. Any disagreements regarding study inclusion were resolved by discussion or decided by a third reviewer (PC) where necessary.

Study Selection and Eligibility Criteria:

Studies were included if they:

(i) were published from April 2011 (the date of the initial KFRE paper publication) to 5th February 2021

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(ii) studied adults with CKD but not receiving renal replacement therapy

(iii) used the KFRE to estimate the risk of ESRD, and

(iv) evaluated the actual rather than theoretical impact of utilising the KFRE in clinical practice.

Studies were excluded if they were development or validation studies only, narrative reviews, editorials, commentaries or opinions, or letters.

The two reviewers (HB and AF) independently screened the titles and abstracts against the eligibility criteria. Potentially relevant studies were identified, and final inclusion was based on full-text examination. Reasons for exclusion following full-text review were documented.

Data Extraction and Quality Assessment:

The following key data were extracted: (i) study details (author, year of publication, title, location of study), (ii) study type, (iii) aim, (iv) study population characteristics, (v) method, (vi) results, (vii) key findings, (viii) strengths and limitations, and (ix) author conclusions.

Methodological validity was assessed independently by the two reviewers (HKB and AF) using the Critical Appraisal Skills Programme (CASP) appraisal tools[14], Joanna Briggs Institute (JBI) Critical Appraisal Tools[15] and Centre for Evidence-Based Management (CEBMa) Critical Appraisal of a Survey.[16]

Data synthesis and Analysis:

Due to different study methods and the heterogeneity of study characteristics, individual analysis for quantitative studies and qualitative studies was not conducted. Findings were

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reported as a narrative synthesis adopting the methods presented by Popay et al.[17]: (i) theoretical reason for basis of evaluated intervention, i.e. using the KFRE, (ii) descriptive summary of study characteristics and critical appraisal, (iii) exploration of associations within and between studies, and (iv) assessing the strength of evidence and limitations of the synthesis process.

Patient and Public Involvement:

There was no patient and public involvement in the conduct of this systematic review.

RESULTS

The database search identified 947 studies, and a further 478 studies were identified from the forward citation search of the KFRE development study.[13] Three hundred and sixty-seven duplicates were removed, resulting in 1058 studies taken forward for the title and abstract screening. Five studies fulfilled the criteria for full-text review. No additional studies were identified from reference review and forward citation review of the selected studies. Results of the search are presented in the (PRISMA) flow diagram[12] (Figure 1).

Study characteristics

Table 1 presents the details of the included studies. Three studies took place in Canada[3,18,19], one in Australia[20] and one in the US.[21] Four studies were quantitative, and one study used a mixed-methods approach, including qualitative (interviews) and quantitative (surveys) elements.[3] The KFRE was used at three transition points: i) the primary to speciality care interface[18,20], ii) general nephrology to multidisciplinary care or advanced care kidney

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clinics[3,19], and iii) to guide treatment planning in a private healthcare setting[21], including referrals to primary or nephrology care, medication changes and laboratory recommendations. No study stated the baseline risk used (North-American or Non-North-American). Only two studies stated which version of the KFRE was used, but all studies stated if they calculated 2-year or 5-year ESRD risk. Results are shown in Table 2, along with key findings from the critical appraisal. The three themes identified are described below.

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Table 1: Characteristics of included studies

Study	Healthcare setting, Country	Study design	Population, N	KFRE details	Inclusion/exclusion criteria	Intervention	Control	End-points	Follow-up
Hingwala et al. (2017)[18]	Primary to nephrology care, Manitoba, Canada	Quasi-experimental	Patients referred to 3 renal centres in the Manitoba Renal Program, n/a	4-variable KFRE for 5-year risk	12-month period immediately following implementation of triage was observed as a transition period to the new triage system and was excluded from the final analysis. No triage unless all 4 variables available or other indication provided.	Referral if significant criteria OR KFRE 5-year risk >3%. If no criteria met then back to referrer with "low risk letter". If risk 3-10% book as non-urgent (< 6 months) If risk >10% book as urgent (< 4 weeks).	Pre-triage - No set criteria. Referrals triaged as urgent, nonurgent, or "do not book" by an individual rotating nephrologist.	Compared between pre and post triage periods: 1. Wait time between referral and nephrology visit (days). 2. Number of consults.	Pre-triage 1st Jan 2011 to 31st Dec 2011. Post-triage 1st Jan 2013 to 31st Dec 2013
Hong et al. (2020)[20]	Primary to nephrology care, New South Wales, Australia	Pre-post	Patients referred to St George Hospital Renal Department, n/a	KFRE for 5-year risk, unclear on number of variables		In Jan 2019 triage consultant and risk-based triage introduced. Referral if KFRE 5-year risk >3%. If < 3% risk, referral on consultant discretion.	Pre-triage periods in 2018 and 2017.	Number of consults between pre- and post- triage periods.	2017 to 2019
Smekal et al. (2019)[3]	Multidisciplinary to general nephrology care, Alberta, Canada	Mixed-methods	Interviews, 27: Southern Alberta Renal Program, 9 patients & 1 family member; multidisciplinary CKD healthcare providers in Calgary, Alberta, 17. Survey in Calgary, Alberta: patients from CKD clinics, 413; CKD healthcare providers, 73.	KFRE for 2-year risk, unclear on number of variables .	Interviews: Patients age >18 years with non-dialysis CKD, discharged from multidisciplinary to general nephrology care & multidisciplinary CKD healthcare providers (nephrologists, nurses, allied health professionals). Survey: All patients from CKD multidisciplinary and general nephrology clinics & all multidisciplinary CKD healthcare providers.	Transition to CKD multidisciplinary clinic when KFRE 2-year risk ≥ 10% or eGFR ≤ 15 ml/min/1.73m^2 implemented in 2017. Interviews post-implementation. Surveys post-implementation: patient – paper-based care experience survey & provider – online job satisfaction survey.	Surveys pre-implementation: patient – paper-based care experience survey & provider – online job satisfaction survey.	Interview data collection stopped once data saturation reached. Survey responses returned within time period.	Survey distributed to patients Nov 2016-Jan 2017 (pre-implementation) and Jan 2018-Mar 2018 (post implementation. Survey responses from providers Nov 2016-Jan 2017 (pre-implementation) and Apr 2018 (post-implementation).
Che et al. (2020)[19]	Discharges from multidisciplinary	Retrospective cohort	Prevalent CKD patients in MCKC in 2013 with	4- variable KFRE for 2-	Included CKD patients from MCKC with available data.	Revised eligibility criteria between 2016 and 2018 from eGFR < 30ml/min/1.73m^2 to		Number discharged from MCKC, re-referred,	2013 to Jan 2020

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	care, Ontario, Canada		available data, 643.	and 5-year risk		eGFR <15ml/min/1.73m^2 & KFRE 2-year risk > 10%. If not eligible then discharged from MCKC.	commenced RRT, and died.	
Sendak et al. (2016)[21]	Private care, North Carolina, US	Prospective cohort	Duke Connected Care a MSSP, 413.	KFRE for 2-year risk, unclear on number of variables	Patients alive and without evidence of ESRD. Excluded if had past nephrology visit and those with AKI without chronic dysfunction.	Patients with a KFRE 2-year risk >15% referred for "population health rounding" – in-depth EHR weekly review with MDT to decide on changes in management	Number of patients rounded per month, time per case during rounds, % of patient at rounds that have action taken, incidence of ESRD, RRT modality, number of dialysis cash-starts.	June 2015 for 5 months

N – number; KFRE – Kidney Failure Risk Equation; CKD – Chronic kidney disease; eGFR – estimated glomerular filtration rate; MCKC – Multi-Care-Kidney-Clinic; RRT – renal replacement therapy; MSSP – Medicare Shared Savings Program; AKI – acute kidney injury; HER – electronic health record; MDT – multidisciplinary team.

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Table 2: Results and critical appraisal of included studies

Study	Baseline characteristics				Results	Key findings	Strengths	Limitations	Additional Points from Critical Appraisal
	Age (years)	Sex (male [%])	eGFR (ml/min/1.73m²)	Urine ACR					
Hingwala et al. (2017)[a]	Pre-triage: not presented Post-triage: 68 (median)	Pre-triage: not presented Post-triage 50.3	Pre-triage: not presented Post-triage: 39 (median)	Pre-triage: not presented Post-triage: 9.2 (median)	Pre-triage: - Referrals booked for appointment: not presented - Median monthly referrals: 68/month (range 44-76) - Median wait time: 230 days (range: 126-355) Post-triage: - Referrals booked for appointment: 66% - Median monthly referrals: 94/month (range 61-147) - Median wait time 58 days (range: 48-69)	Monthly referral increase of 45%. Median wait times improved (P < 0.001)	Low-cost triage system Triage process allows for clinical judgment, such that 'low-risk' patients according to KFRE are identified for nephrology review where necessary	Lack of data presented for pre-triage period (e.g baseline characteristics, number of booking) to allow for detailed historical comparison (pre-triage) No clinical or economic impact assessment of implementing KFRE triage system No long-term follow-up on patients triaged as low risk. May not be applicable to teams operating under capacity without wait lists.	Washout period for 1 year when implementing criteria Additional capacity to see referrals (additional nephrologist hired in Sept 2013, physician-assistant led clinic from Sept 2012) "Hawthorne effect" 3 centres No financial support. Authors declared no competing interests.
Hong et al. (2020)[a]	Not presented	Not presented	Not presented	Not presented	Compared to 2018 when KFRE-based triage implemented: - 25% fewer consults in 2019 (30% less than in 2017) - Fewer patients had a low-risk KFRE at triage (46% vs. 48%) - Fewer low-risk patients had clinic follow-up (50% vs. 52%) More low-risk patients remaining in clinic (86% vs. 60%) had alternative reasons for follow-up (i.e. eGFR < 30 mL/min, moderate proteinuria, or uncontrolled hypertension)	Implementing the KFRE based triage system reduced overall and low-risk patient numbers in outpatient clinics.	Triage system allowed for consultant discretion for low-risk patients to be followed up	No baseline characteristics presented for patients either pre or post implementation of the KFRE based triage process Low-risk patients with eGFR < 30 or moderate proteinuria still followed up Inability to calculate KFRE score for referrals at triage (28% in 2018, 36% in 2019) due to missing urine ACR	Limited data therefore unable to compare if pre- vs post-triage groups were similar Single centre Unclear if results "significant" No comment on what was done for patients with missing ACR

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Smekal et al. (2019)[b,c]	Pre-KFRE implementation: Patients: - survey 10% < 50; 28% 50-64; 22% 65-74; 39% ≥ 75. Healthcare providers: not presented. Post-KFRE implementation: Patients: - interviews 10% 50-64; 30% 65-74; 60% ≥ 75. - survey 17% < 50; 19% 50-64; 27% 65-74; 38% ≥ 75. Healthcare providers: not presented.	Pre-KFRE implementation: - Patients (survey): 60. - Healthcare providers (survey): 28. Post-KFRE implementation: Patients: - interviews 50 - survey 57. Healthcare providers - interviews 29 - survey 27.	Not presented	Not presented	Interviews: - 9/23 (39%) patients & 17/75 (23%) providers interviewed. - Five themes were identified among patients and providers and two additional categories identified among providers only. Questionnaire: - Majority of patients satisfied with their care in both periods with no overall differences. However, there were improvements in patients' experience of access to care, caring staff, and safety of care. - Of the 75 providers 40 (53%) and 33 (44%) completed the pre- and post-implementation job satisfaction survey, respectively; no differences in providers' job satisfaction.	Patients and healthcare providers reported - improved the focus of MDT clinics by targeting high-risk patients - using KFRE to target care to high-risk patients was a key strength. Enhanced the sustainability of the clinics Providers expressed concern that there may be inadequate access to and lower quality of care for low-risk patients discharged from multidisciplinary care, although patients did not experience this. No difference in patients' care experience. No difference in healthcare provider job satisfaction.	Includes both patient and provider perspectives. Presents both qualitative and quantitative data.	More healthcare providers than patients in qualitative component; data saturation was achieved in both groups. Most patients interviewed had >5 years in MDT care prior to discharge so may not be representative of general nephrology patients. Patients discharged from MDT clinics within previous 12 months so limited time period following implementation of risk approach. Unable to pair pre- and post-survey responses or establish response rate. Single centre.	Limited to English speaking participants. Low response rate. Detailed information on data collection and analysis, questions provided. Mostly female providers. Funder had no role in the project. Authors declared no competing interests.
Che et al. (2020)[d]	Not presented	Not presented	< 30	Not presented	470 (73%) continued follow-up in MCKC. Of 142 (22%) discharged to primary care: - 52 (37%) died - 15 (11%) re-referred to nephrology (at median 982 [IQR 560] days) - 8 (6%) initiated RRT (at median 850 [1411]) days; 5 [63%] for unforeseen acute illness) 31 (5%) discharged to general nephrology.	Discharge of a significant number of patients when moving to new criteria, few of whom ultimately required RRT that could have been prevented.	"Low loss to follow-up".	Completeness of data and follow-up unclear. May not be generalisable as based on one regional renal program.	Unclear reason for chosen threshold or if this is the ideal level. Limited data No comparison or control group. No detail on missing data
Sendak et	Not presented	Not presented	Not presented	Not	Of 335 patients of 413 eligible:	Patients with CKD at	KFRE easily applicable	No data on long term	Elderly patients

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al. (2016)[d]		presented	73 (21.8 %) management changes: - 53 (72.6%) to nephrology - 8 (11.0%) to primary care - 7 (9.6%) lab recommendations - 4 (5.5%) medication recommendations. Average time-per-case per health round 2m 12 s. Of the remaining 262 (78.2%) patient screening did, however, identify; - 110 (42.0%) seeing a nephrologist - 35 (13.3%) recently deceased - 25 (9.5%) on dialysis.	high risk of progression to ESRD can be identified using validated algorithms applied to structured data that is readily available. A significant proportion of patients identified in this way require management changes, including patients that require nephrology review. Limited ability to screen out patients using structured data. Health rounds can be performed relatively rapidly.	Highlights additional patients for review in primary/speciality care.	outcomes. Data from one renal program, therefore, may not be generalisable.	Later months more efficient as workflow optimised – not taken into consideration for time per case. Private healthcare setting so not generalisable.

eGFR – estimated glomerular filtration rate; ACR – albumin:creatinine ratio; KFRE – Kidney Failure Risk Equation; MDT – Multidisciplinary team; MCKC – Multi-Care-Kidney-Clinic; RRT – renal replacement therapy; CKD – chronic kidney disease; ESRD – end-stage renal disease.

Appraisal tool used for the studies:

[a] Joanna Briggs Institute Quasi-experimental study checklist

[b] CASP Qualitative Studies Checklist (interviews)

[c] Centre for evidence-based management critical appraisal of a survey

[d] CASP Cohort Study Checklist

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Primary care to speciality care interface

Two studies[18,20] reported outcomes following the introduction of a new referral process incorporating a 5-year ESRD risk threshold of >3% to triage patients from primary care to speciality care; although they also incorporated additional referral criteria for eligibility for speciality care review. Both studies measured the number of consults pre- and post-implementation of the KFRE: Hingwala et al. found monthly referrals increased by 45%[18], whereas Hong et al. found referrals decreased by 25-30%.[20]

Hingwala et al. also showed a significant reduction in the wait-time from referral to review (median 58 vs 230 days) following implementation of the new triage system.[18]

General nephrology to multidisciplinary/advanced kidney care clinic

Patients under nephrology care may be managed in a general nephrology clinic or, for patients with more advanced CKD, a multidisciplinary clinic focused on managing CKD complications and RRT preparation. Two studies examined the use of the KFRE at this interface and introduced similar eligibility criteria for entry into and management in the multidisciplinary clinic: a 2-year ESRD risk > 10% or eGFR < 15 ml/min/1.73m^2.

Che et al.[19] found that by applying these criteria to all prevalent patients in the multidisciplinary clinic, 73% remained in the clinic, 5% were stepped down to general nephrology care, and 22% were discharged to primary care. Of the latter, 11% required re-referral to nephrology, and 6% ultimately required RRT, although the majority in the context of an unforeseen acute illness.

Smekal et al.[3] evaluated patient and healthcare providers' views and experiences following the implementation of the new KFRE-based criteria for multidisciplinary clinic management.

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Providers felt using KFRE to target high-risk patients was a key strength, limiting inappropriate referrals and improving the clinic's focus. The caseload was felt to be more 'acute', but overall workload not significantly changed, and there was no significant difference in job satisfaction. Providers also expressed concern that there may be inadequate access to and a lower quality of care for low-risk patients discharged from the multidisciplinary clinic, such as with education and monitoring. Some nurses reported discharged patients were contacting them for test results, and patients reported missing this aspect of their care. Although there was some improvement in patient's experience of access to care, caring staff, and safety of care, most patients were satisfied with their care both pre- and post-implementation, and there was no difference in patients' overall care experience.

Treatment planning

The study by Sendak et al.[21] took place in an "Accountable Care Organisation" within the Medicare Shared Savings Program (Medicare is a health insurance program mainly for people aged ≥65 years in the US). A 2-year ESRD risk threshold of >15% was applied to the cohort to identify high-risk CKD patients, not already under nephrology review, for a multidisciplinary review of their electronic healthcare record (EHR). The reviews were able to be performed relatively rapidly (average 2 minutes 12 seconds) and led to changes in management in 21.8% of cases, most often a referral for nephrology review.

Critical appraisal

All studies were limited to a single centre or region. No randomised control trials were identified. Two were cohort studies[19,21], one was a mixed-method study[3] and two were quasi-experiment studies[18,20]. All studies stated a clear aim or issue to be addressed. Sample size varied depending on the study method, and two studies did not state it.[18,20] Participant

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baseline characteristics data was not available in two studies[19,20] and lacking in one study[18] where pre-intervention information was not available. For the qualitative aspect of the mixed-methods study[3], the patients selected were English-speaking only. Although this may have allowed for simpler data analysis as there was no requirement for language translation, this selection bias may limit the perspectives obtained of those experiencing the intervention. In this same study, the survey response rate for patients was unclear, and for providers, no data was provided to establish any difference in demographics to non-responders. Although mostly proportions were reported in the data analysis of quantitative studies, there was a lack of reporting of statistical significance, strength of any associations or measures of precision in most. Three studies[19-21] reported only proportions when applying the intervention. These three studies had not commented on the number of patients, if any, who could not be assessed for risk due to missing data or whether there were any confounding factors.

DISCUSSION

In this systematic review, we identified five studies of varying methodologies evaluating the actual impact of utilising the KFRE in clinical practice for patients with CKD. The equation was utilised in three main areas: i) triaging patients between primary and speciality nephrology care, ii) directing patients between general nephrology and multidisciplinary advanced CKD care clinics, and iii) treatment planning where high-risk patients within a healthcare program were identified for a multidisciplinary EHR review.

Interpretation of the results in the context of other evidence

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Previous studies exploring the potential impact of applying the KFRE at the primary care-speciality care interface suggested it could increase referral numbers.[5,22,23] This is consistent with the findings of Hingwala et al.[18], who reported an increase in referrals. However, wait times improved, possibly related to additional capacity to see referrals and a "Hawthorne effect" (better performance as a result of health care professionals' awareness of being observed in a study).[24] In contrast, Hong et al.[20], who implemented the same risk threshold, reported a fall in referrals. These contrasting findings likely reflect differences in local practice and case-mix, whereby a higher proportion of low-risk patients (i.e. older, higher eGFRs, lower urine ACRs) were managed by nephrology rather than by primary care before implementation of the KFRE criteria, compared to the centre reported by Hingwala et al. We were unable to review any population differences pre- and post- triage or between studies due to a lack of reported data.[18,20]

In specialist nephrology care, many centres have advanced kidney care clinics for those patients who require increased multidisciplinary support to manage CKD complications and prepare for RRT.[25,26] The KFRE has the potential to identify patients at higher risk of ESRD to more efficiently direct increased support and resources for this group. Those identified as low risk can be stepped down, reducing clinic burden and unnecessary interventions. Two studies evaluated this part of the CKD pathway and applied similar eligibility criteria for multidisciplinary clinic management. Che et al.[19] found many patients could be discharged from the multidisciplinary clinic, the majority back to primary care, with only a small number referred back to nephrology care or ultimately requiring RRT.

The findings of Smekal et al.[3] suggest some anxiety regarding reduced access to services, education and monitoring when patients are discharged from the multidisciplinary clinic to

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general nephrology care based on their KFRE-calculated risk. Maintaining patient and provider satisfaction with CKD care when using the KFRE to discharge patients is an important issue that will require the considered configuration of local renal service delivery to ensure accessibility and patient safety. It remains unknown if the application of the KFRE to identify higher-risk patients for multidisciplinary care impacts key outcomes such as progression of CKD, commencing RRT, cardiovascular events or mortality.

Sendak et al.[21] reviewed the impact of utilising KFRE in a private healthcare setting in a primarily elderly population to identify patients who required input from medical teams (in primary or specialist care), treatment changes or additional lab testing. The results of this study may not be easily generalisable as this is not a widely used provider model in some countries.

Other suggested areas of use have been for planning RRT in an elderly population where competing risk of death is of concern[27] and for creation of vascular access[28] but no studies have implemented and evaluated this.

Limitations of the evidence included in the review and the review process

The studies varied widely in methodologies used. The benefit of this mixed-methods review is a greater breadth of understanding and evidence around the application of the risk-based KFRE in clinical practice. This can provide a more rounded body of evidence to inform changes in clinical practice and policy decision making. No randomised control trials were identified.

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Study quality also ranged widely. Some studies did not provide adequate detail of their population characteristics. As a result, it was difficult to assess if the population was representative or findings could be generalised. Recognising differences between study populations was also limited by the lack of data. Few studies adequately completed statistical analysis and so it was challenging to decipher significant findings.

Implication of the results for practice, policy, and future research

Whilst this study has identified the ease of use of the KFRE to triage patients and highlight those who would benefit from changes in management, plus the potential effect on the number of referrals and patients' and providers' experience and perspectives, the impact on health outcomes and economic benefit are still unknown.

More high-quality studies are needed to confidently support the widespread use of KFRE, particularly randomised-control trials with a focus on health outcomes and economic impact. Four trials that are in currently progress will assess the outcomes of applying the KFRE. The Kidney CHAMP study[29], a cluster randomised controlled trial in Pittsburgh, US, will review the effectiveness of integrating the KFRE into EHR to identify patients at high risk of progression who require intervention such as nephrology guidance, medication reviews and targeted CKD education. Outcomes assessed will be a composite of a 40% reduction in eGFR and ESRD. A multicentre cluster randomised control trial in Canada[30] aims to review the risk-based approach in the primary care setting. The intervention, providing patients and providers in primary care with a patient's KFRE score and the risk-based referral criteria, will be reviewed to assess if it improves appropriate patient management, improves a patient's CKD-specific health literacy and affects the cost of care compared with usual care. The PREPARE NOW study[31], a cluster randomised control trial within a nephrology speciality

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care setting in Pennsylvania, US, will be applying the KFRE to the intervention group to alert the health care team of patients at risk of progression (10% 2-year risk) and in need for intervention. This will be in addition to multiple other components, and patient-reported, biomedical and health system outcomes will be collected. We presented one of the three studies in this systematic review[3] resulting from a multiphase mixed-methods study.[32] The whole study took on a pre-post design and applied the KFRE to patients in nephrology multidisciplinary CKD clinics in Alberta, Canada. The final phase will aim to evaluate the costs of care and outcomes before and after the introduction of risk-based triage, such as health care resource use, frequency of testing, modality choice, and death.

Despite the lack of sufficient impact studies, strong evidence to date from validation studies and studies investigating potential impact in clinical practice have been encouraging. As a result, several CKD guidelines have or are in the process of incorporating KFRE risk-based criteria in their pathways.[6,7,33]

Conclusion

The KFRE has been extensively validated, but there has been relatively little evaluation of its clinical and economic impact. This is the first study to systematically review studies exploring at the actual impact of using the KFRE in clinical practice. Additional high-quality studies are required, and trials assessing the impact of using KFRE at various stages across the CKD pathway are in progress.

ACKNOWLEDGEMENTS

April Coombe – Information specialist

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COMPETING INTERESTS

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The views expressed are those of the author(s) and not necessarily those of the NIHR or the Department of Health and Social Care.

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AUTHOR CONTRIBUTIONS

HKB, PC, MC and DK conceived the study. HKB developed the methodology with guidance from PC, MC, DK, OLA and AF. The database, reference and citation searches were completed by HKB and AF. Screening and selection of the included studies was conducted by HKB and AF; PC as the third reviewer was consulted for any disagreements. HKB and AF critically appraised the included studies. HKB drafted the initial manuscript and all authors contributed to revision and approval of the final manuscript.

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REFERENCES

1. Bikbov B, Purcell CA, Levey AS, et al. Global, regional, and national burden of chronic kidney disease, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *The Lancet.* 2020;395(10225):709-733.
2. Daker-White G, Rogers A, Kennedy A, Blakeman T, Blickem C, Chew-Graham C. Non-disclosure of chronic kidney disease in primary care and the limits of instrumental rationality in chronic illness self-management. *Social science & medicine (1982).* 2015;131:31-39.
3. Smekal MD, Tam-Tham H, Finlay J, et al. Patient and provider experience and perspectives of a risk-based approach to multidisciplinary chronic kidney disease care: A mixed methods study. *BMC Nephrology.* 2019;20(1).
4. Chronic kidney disease in adults: assessment and management (CG182). London, National Institute for Health and Care Excellence (UK), 2014.
5. Bhachu H, Cockwell P, Subramanian A, et al. Impact of utilising risk-based stratification on referral of patients with chronic kidney disease from primary care to specialist care in the UK. *Kidney International Reports.* 2021;Article in press.
6. Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group: KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Kidney Inter suppl, 3**:** 1-150, 2013.
7. Farrington K, Covic A, Aucella F, et al. Clinical Practice Guideline on management of older patients with chronic kidney disease stage 3b or higher (eGFR <45 mL/min/1.73 m2). *Nephrology Dialysis Transplantation.* 2016;31(suppl_2):ii1-ii66.

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8. Ramspek CL, de Jong Y, Dekker FW, van Diepen M. Towards the best kidney failure prediction tool: a systematic review and selection aid. *Nephrol Dial Transplant.* 2020;35(9):1527-1538.

9. Tangri N, Grams ME, Levey AS, et al. Multinational Assessment of Accuracy of Equations for Predicting Risk of Kidney Failure: A Meta-analysis. *JAMA.* 2016;315(2):164-174.

10. Ramspek C, De Jong Y, Dekker F, Van Diepen M. Predictive models for progression of chronic kidney disease to kidney failure: A systematic review. *Nephrology Dialysis Transplantation.* 2018;33 (Supplement 1):i155.

11. The Kidney Failure Risk Equation. Accessed 30/10/2017.

12. Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. *PLoS Med.* 2009;6(7):e1000100.

13. Tangri N, Stevens LA, Griffith J, et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA.* 2011;305(15):1553-1559.

14. Critical Appraisal Skills Programme. https://casp-uk.net/. Accessed 14/4/2021.

15. Joanna Briggs Institute Critical Appraisal Tools. https://jbi.global/critical-appraisal-tools. Accessed 14/4/2021.

16. Critical Appraisal of a Survey. Center for Evidence-Based Management (CEBMa). https://www.cebma.org/wp-content/uploads/Critical-Appraisal-Questions-for-a-Survey.pdf. Accessed 14/4/2021.

17. Popay J, Roberts H, Sowden A, et al. Guidance on the Conduct of Narrative Synthesis in Systematic Reviews: A Product from the ESRC Methods Programme. 2006.

For peer review only

18. Hingwala J, Wojciechowski P, Hiebert B, et al. Risk-Based Triage for Nephrology Referrals Using the Kidney Failure Risk Equation. *Can J Kidney Health Dis.* 2017;4:2054358117722782.

19. Che M, Thanabalasingam SJ, Iliescu EA, White CA. Patient outcomes following discharge from a CKD clinic. *Journal of the American Society of Nephrology.* 2020;31:227.

20. Hong R, Pirabhahar S, Turner K, Katz IJ. Triage system for nephrology referrals using the kidney failure risk equation (KFRE) score. *Nephrology.* 2020;25 (SUPPL 3):53.

21. Sendak M, Cameron B, Komives E, et al. Developing a data-driven work flow for population health rounding. *Journal of General Internal Medicine.* 2016;31(2):S874-S875.

22. Slattery LM. Applying novel kidney failure risk equations applications to new patient referrals: A 6-month retrospective review. *Journal of the American Society of Nephrology.* 2019;30:1173.

23. Major RW, Shepherd D, Medcalf JF, Xu G, Gray LJ, Brunskill NJ. The kidney failure risk equation for prediction of end stage renal disease in UK primary care: An external validation and clinical impact projection cohort study. *PLoS Medicine.* 2019;16(11).

24. McCambridge J, Witton J, Elbourne DR. Systematic review of the Hawthorne effect: new concepts are needed to study research participation effects. *J Clin Epidemiol.* 2014;67(3):267-277.

25. Nicoll R, Robertson L, Gemmell E, Sharma P, Black C, Marks A. Models of care for chronic kidney disease: A systematic review. *Nephrology (Carlton).* 2018;23(5):389-396.

For peer review only

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26. Evans M, Lopau K. The transition clinic in chronic kidney disease care. *Nephrol Dial Transplant.* 2020;35(Suppl 2):ii4-ii10.

27. Nastasa A, Apetrii M, Onofriescu M, et al. Risk prediction for death and end-stage renal disease does not parallel real-life trajectory of older patients with advanced chronic kidney disease-a Romanian center experience. *Nephrology Dialysis Transplantation.* 2020;35 (SUPPL 3):iii486.

28. Inston N, Lok CE. Improving precision in prediction: Using kidney failure risk equations as a potential adjunct to vascular access planning. *Journal of Vascular Access.* 2019;20(1):95-97.

29. Jhamb M, Yabes J, Fischer G, et al. Electronic health record based population health management for improving CKD care: the kidney-coordinated health management partnership (CHAMP) study. *Journal of the American Society of Nephrology : JASN.* 2019;30:525-.

30. Harasemiw O, Drummond N, Singer A, et al. Integrating Risk-Based Care for Patients With Chronic Kidney Disease in the Community: Study Protocol for a Cluster Randomized Trial. *Canadian Journal of Kidney Health and Disease.* 2019;6(no pagination).

31. Green JA, Ephraim PL, Hill-Briggs FF, et al. Putting patients at the center of kidney care transitions: PREPARE NOW, a cluster randomized controlled trial. *Contemp Clin Trials.* 2018;73:98-110.

32. Hemmelgarn BR, Smekal MD, Weaver RG, et al. Implementation and evaluation of a risk-based approach to guide chronic kidney disease care: Protocol for a multiphase mixed-methods study. *Canadian Journal of Kidney Health and Disease.* 2018;5(no pagination).

For peer review only

33. Chronic kidney disease: assessment and management (update). In development [GID-NG10118]. https://www.nice.org.uk/guidance/indevelopment/gid-ng10118/documents. Accessed 14/6/2021.

Figure 1: Flow diagram to show the studies identified from searches

(See separate file – "FIGURE 1")

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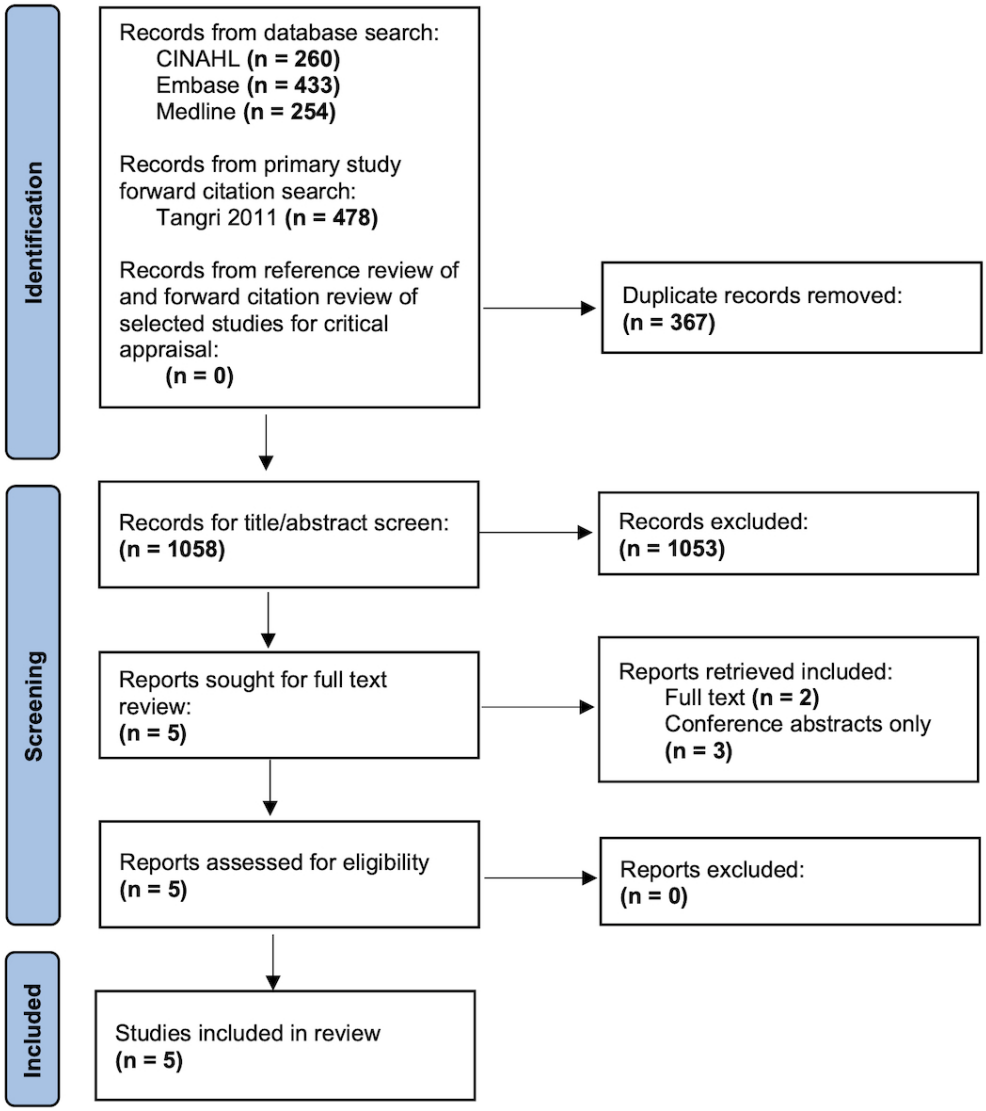
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SUPPLEMENTARY MATERIAL TABLE OF CONTENTS

Table S1: Search strategies for the searches

Table S2: Prisma Checklist

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Flow diagram to show the studies identified from searches

90x104mm (300 x 300 DPI)

SUPPLEMENTARY MATERIAL

Table S1: Search strategies for the searches

Database	Search strategy/ search terms	Results
MEDLINE (Ovid)* In-Process & Other Non-Indexed Citations 1946 to February 05, 2021	1. (CKD or CRF or CRD or CKF).ti,ab.	48554
	2. (renal insufficiency or kidney failure or kidney disease* or renal disease* or renal failure).ti,ab.	229317
	3. exp Renal insufficiency, Chronic/	117932
	4. 1 or 2 or 3	288873
	5. (kidney failure risk* or KFRE).ti,ab.	76
	6. 4 and 5	76
	7. limit 6 to yr="2011 -Current"	71
	8. (kidney failure adj3 risk*).ti,ab.	312
	9. KFRE.ti,ab.	26
	10. 8 or 9	312
	11. 4 and 10	312
	12. Limit 11 to yr="2011 -Current"	254
Embase (Ovid)* 1974 to 2021 February 04	1. (CKD or CRF or CRD or CKF).ti,ab. [84422]	84422
	2. (renal insufficiency or kidney failure or kidney disease* or renal disease* or renal failure).ti,ab.	342667
	3. exp chronic kidney failure/	106053
	4. 1 or 2 or 3	403062
	5. (kidney failure risk* or KFRE).ti,ab.	139
	6. 4 and 5	139
	7. limit 6 to yr="2011 -Current"	135
	8. (kidney failure adj3 risk*).ti,ab.	498
	9. KFRE.ti,ab.	61
	10. 8 or 9	499
	11. 4 and 10	499
	12. Limit 11 to yr="2011 -Current"	433
CINAHL Plus* (EBSCOhost)	1. TI (CKD OR CKF OR CRD OR CRF) OR AB (CKD OR CKF OR CRD OR CRF) [11577]	11577
	2. TI (renal insufficiency OR kidney failure OR kidney disease* OR renal disease* OR renal failure) OR AB (renal insufficiency OR kidney failure OR kidney disease* OR renal disease* OR renal failure) [51205]	51205
	3. (MH "Renal Insufficiency, Chronic+") [28518]	28518
	4. S1 OR S2 OR S3 [65333]	65333
	5. TI KFRE OR AB KFRE [5]	5
	6. TI (kidney failure risk*) OR AB (kidney failure risk*) [335]	335
	7. S5 OR S6 [335]	335
	8. S4 AND S7 [335]	335

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	9. S4 AND S7 Limiters - Publication Year: 2011-2021[260]	260
Web of Science*	Tangri 2011 A predictive model for progression of chronic kidney disease to kidney failure	478
Forward citation review	Selected studies	0
Reference review	Selected studies	0

*Search date 05/02/2021

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Table S2: Prisma Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Pages 2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 4
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 4
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Pages 5-6
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 5, Table S1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table S1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Pages 5-6
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 6
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 6
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 6
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 7
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 7
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 7
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 7

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Section and Topic	Item #	Checklist item	Location where item is reported
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	NA
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 7
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 7
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 7
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 7, Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Pages 7-9, Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Page 10, Table 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 2
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Table 2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Table 2
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 10
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 10
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 10
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Pages 10-12
	23b	Discuss any limitations of the evidence included in the review.	Pages 12-13

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Section and Topic	Item #	Checklist item	Location where item is reported
	23c	Discuss any limitations of the review processes used.	Pages 12-13
	23d	Discuss implications of the results for practice, policy, and future research.	Pages 13-14
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Pages 3 & 5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Pages 3 & 5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 16
Competing interests	26	Declare any competing interests of review authors.	Pages 15-16
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA, Data in Tables 1&2

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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Prisma Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
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	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 10
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 10
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Pages 10-12
	23b	Discuss any limitations of the evidence included in the review.	Pages 12-13

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Section and Topic	Item #	Checklist item	Location where item is reported
	23c	Discuss any limitations of the review processes used.	Pages 12-13
	23d	Discuss implications of the results for practice, policy, and future research.	Pages 13-14
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Pages 3 & 5
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Pages 3 & 5
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 16
Competing interests	26	Declare any competing interests of review authors.	Pages 15-16
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA, Data in Tables 1&2

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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USE OF THE KIDNEY FAILURE RISK EQUATION TO INFORM CLINICAL CARE OF PATIENTS WITH CHRONIC KIDNEY DISEASE: A MIXED-METHODS SYSTEMATIC REVIEW

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USE OF THE KIDNEY FAILURE RISK EQUATION TO INFORM CLINICAL CARE OF PATIENTS WITH CHRONIC KIDNEY DISEASE: A MIXED-METHODS SYSTEMATIC REVIEW

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ABSTRACT

Rationale & Objective: The Kidney Failure Risk Equation (KFRE) predicts the risk of end-stage kidney disease in patients with chronic kidney disease (CKD). This study aimed to evaluate the impact of the utility of KFRE in clinical practice.

Study Design: Systematic review.

Setting & Study Populations: Adult patients with CKD but not receiving renal replacement therapy enrolled in studies where KFRE was used in clinical care pathways.

Selection Criteria for Studies: All studies published from April 2011 to October 2021 identified from Medline, CINAHL, Embase and reference and citation searches of included studies.

Data Extraction: Relevant data were extracted, and two reviewers independently assessed study quality using appropriate appraisal tools.

Analytical Approach: Findings reported as a narrative synthesis due to heterogeneity of the included studies.

Results: Of 1635 studies identified, 440 duplicates were removed. The remaining 1195 titles and abstracts were screened. All five studies for full-text review were included in the analysis. Three uses of KFRE were assessed: i) primary to speciality care interface; ii) general nephrology to multidisciplinary care transition; iii) treatment planning. Evidence of

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impact on number of patient referrals into nephrology care was conflicting. However, wait times improved in one study. Although KFRE identified high-risk patients for increased multidisciplinary support, there was concern patients stepped down, no longer meeting eligibility criteria, may lack access to services.

Conclusions: This is the first systematic review of studies that have assessed the actual impact of KFRE in clinical practice with five studies of varying quality reported to date. Trials are in progress assessing the impact on clinical outcomes of using KFRE in clinical practice, and KFRE is being incorporated into guidelines for CKD management. Further studies are needed to assess the impact of KFRE on clinical care.

Registration: Protocol registered on PROSPERO before initiation of the study (Ref: CRD42020219926).

DATA AVAILABILITY STATEMENT

Data sharing not applicable as no datasets generated and/or analysed for this study.

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STRENGTHS AND LIMITATIONS OF THIS STUDY

- To the best of our knowledge, this is the first systematic review to examine the evidence for the impact of the use of the Kidney Failure Risk Equation in clinical practice.
- A mixed-methods approach was used to allow the inclusion of both quantitative and qualitative evidence.
- Study quality ranged widely, and some studies did not provide adequate detail of their population characteristics, such that generalisability was difficult to assess.
- Further, the statistical analysis was limited in several studies.

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BACKGROUND

The global prevalence of chronic kidney disease (CKD) is estimated to be 9.1% and CKD was the 12th leading cause of death in 2017.[1] CKD is also associated with an increased risk of progression to end-stage kidney disease (ESKD). Risk stratification in CKD can enable more efficient care, with specialty care targeted to patients at higher risk of ESKD, whilst sparing those with low risk unnecessary intervention and undue anxiety associated with this.[2,3]

Most guideline criteria for referral of patients to specialist nephrology care use estimated glomerular filtration rate (eGFR), eGFR decline, and urine albumin:creatinine ratio (ACR) thresholds rather than a quantified ESKD risk.[4] Such criteria often result in the referral of patients at low risk of ESKD and the non-referral of patients at high risk.[5] Therefore, recent CKD guidelines endorse risk-based thresholds for specialty referral and renal replacement therapy (RRT) planning.[6,7]

The best validated risk prediction model is the Kidney Failure Risk Equation (KFRE), which predicts the 2- and 5-year risk of ESKD in patients with stages 3-5 CKD and has undergone extensive validation.[8-10] The equation can easily be embedded into electronic medical records and is also readily available online.[11]

It is uncertain whether the use of the KFRE in clinical practice has a meaningful impact on clinical pathways and health outcomes and how patients and healthcare professionals view the KFRE. We carried out a systematic review of the available evidence of the impact of the use of the KFRE in clinical practice.

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METHODS

The study protocol was registered on PROSPERO (Ref: CRD42020219926), and the study is reported as per the 'Preferred Reporting Items for Systematic Review and Meta-Analysis' (PRISMA) checklist.[12]

Data Sources and Searches:

Two reviewers (HKB and AF) independently searched MEDLINE (Ovid), Cumulative Index to Nursing and Allied Health Literature (CINAHL) (EBSCO) and Embase (Ovid) for studies between April 2011 to October 2021 that examined the impact of utilising the KFRE in patients with CKD.

The search strategies (Table S1) were developed with the support of an information specialist and used keywords, index terms and MeSH terms tailored and applied to each individual database. No language restrictions were applied. We also hand searched the reference lists of included studies and performed a forward citation review of these studies and the original KFRE development study[13]. All results obtained at each stage of the process were entered into EndNote version X9.3.3 (Clarivate Analytics, PA, USA), and duplicates were removed. Any disagreements regarding study inclusion were resolved by discussion or decided by a third reviewer (PC) where necessary.

Study Selection and Eligibility Criteria:

Studies were included if they:

(i) were published from April 2011 (the date of the initial KFRE paper publication) to October 2021

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(ii) studied adults with CKD but not receiving renal replacement therapy (dialysis or kidney transplantation)

(iii) used the KFRE to estimate the risk of ESKD, and

(iv) evaluated the actual rather than potential impact of utilising the KFRE in clinical practice.

Studies were excluded if they were development or validation studies only, narrative reviews, editorials, commentaries or opinions, or letters.

The two reviewers (HB and AF) independently screened the titles and abstracts against the eligibility criteria. Potentially relevant studies were identified, and final inclusion was based on full-text examination. Reasons for exclusion following full-text review were documented.

Data Extraction and Quality Assessment:

The following key data were extracted: (i) study details (author, year of publication, title, location of study), (ii) study type, (iii) aim, (iv) study population characteristics, (v) method, (vi) results, (vii) key findings, (viii) strengths and limitations, and (ix) author conclusions.

Methodological validity was assessed independently by the two reviewers (HKB and AF) using the Critical Appraisal Skills Programme (CASP) appraisal tools[14], Joanna Briggs Institute (JBI) Critical Appraisal Tools[15] and Centre for Evidence-Based Management (CEBMa) Critical Appraisal of a Survey.[16]

Data synthesis and Analysis:

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Due to different study methods and the heterogeneity of study characteristics, individual analysis for quantitative studies and qualitative studies was not conducted. Findings were reported as a narrative synthesis adopting the methods presented by Popay et al.[17]: (i) theoretical reason for basis of evaluated intervention, i.e. using the KFRE, (ii) descriptive summary of study characteristics and critical appraisal, (iii) exploration of associations within and between studies, and (iv) assessing the strength of evidence and limitations of the synthesis process.

Patient and Public Involvement:

There was no patient and public involvement in the conduct of this systematic review.

RESULTS

The database search identified 1099 studies, and a further 536 studies were identified from the forward citation search of the KFRE development study.[13] Four hundred and forty duplicates were removed, resulting in 1195 studies taken forward for the title and abstract screening. Five studies fulfilled the criteria for full-text review. No additional studies were identified from reference review and forward citation review of the selected studies. Results of the search are presented in the (PRISMA) flow diagram[12] (Figure 1).

Study characteristics

Table 1 presents the details of the included studies. Three studies took place in Canada[3,18,19], one in Australia[20] and one in the US.[21] Four studies were quantitative, and one study used a mixed-methods approach, whereby the strengths of both qualitative (interviews) and

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quantitative (surveys) research elements were combined to gain a broader view of their experience applying the KFRE.[3] The KFRE was used at three transition points: i) the primary to speciality care interface[18,20], ii) general nephrology to multidisciplinary care or advanced care kidney clinics[3,19], and iii) to guide treatment planning in a private healthcare setting[21], including referrals to primary or nephrology care, medication changes and laboratory recommendations. No study stated the baseline risk used (North-American or Non-North-American). Only two studies stated which version of the KFRE was used, but all studies stated if they calculated 2-year or 5-year ESKD risk.

Hingwala et al.[18] applied a quasi-experimental (pre-post) study design to patients referred to three renal centres in the Manitoba Renal Program, Manitoba, Canada. Wait times (between referral and nephrology visit) and number of consults were compared from the periods pre- and post-implementation of the new triage criteria: significant criteria or 4-variable KFRE 5-year risk >3%. Patients with a 3-10% risk were booked as non-urgent (< 6 months) and those with a >10% risk were booked as urgent (< 4 weeks).

A similar pre-post study design was used in the study by Hong et al.[20] where they implemented new criteria, using a 5-year KFRE score of >3%, for patients referred to the St George Hospital Renal Department (New South Wales, Canada). Patients referred with a <3% risk were accepted at the discretion of a nephrology consultant. The number of consults pre- and post-triage periods were compared.

The KFRE was applied within a secondary care setting in two studies. Smekal et al.[3] used a mixed-methods approach to gauge the views of patients and providers one year after implementing the KFRE in 2017 to guide transition into the CKD multidisciplinary clinic.

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Criteria for transition were KFRE 2-year risk ≥ 10% or eGFR ≤ 15 ml/min/1.73m^2. Post-implementation interviews took place with 9 patients discharged from the CKD multidisciplinary clinic, the relative of 1 patient discharged from the clinic, and 17 multidisciplinary CKD healthcare providers. Pre- and post-implementation of the new criteria all patients from the CKD multidisciplinary and general nephrology clinics were invited to complete a paper-based experience survey and all multidisciplinary CKD healthcare providers were asked to complete an online job satisfaction survey.

Che et al.[19] retrospectively compared the outcomes of 643 patients from the Multi-Care-Kidney-Clinic, for patients with advanced CKD, before and after the revision of the clinic eligibility criteria, in Ontario, Canada. The new criteria applied included eGFR <15ml/min/1.73m^2 and KFRE 2-year risk > 10%. If eligibility criteria were not met, patients were discharged from the clinic.

Lastly, Sendek et al.[21] applied the 4-variable KFRE (2-year risk >15%) to the population in Duke Connected Care, a Medicare Shared Savings Program that manages the healthcare of over 46,000 Medicare patients. Patients alive and with no evidence of ESKD, a prior nephrology visit or acute kidney injury without chronic dysfunction were referred for "population health rounding" where their electronic health record was reviewed weekly with a multidisciplinary team to decide on changes in management. Number of patients rounded each month, time per case, and any actions taken were recorded.

Results are shown in Table 2, along with key findings from the critical appraisal. The three themes identified are described below.

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Table 1: Characteristics of included studies

Study	Healthcare setting, Country	Study design	Population, N	KFRE details	Inclusion/exclusion criteria	Intervention	Control	End-points	Follow-up
Hingwala et al. (2017)[18 a]	Primary to nephrology care, Manitoba, Canada	Quasi-experimental	Patients referred to 3 renal centres in the Manitoba Renal Program, n/a	4-variable KFRE for 5-year risk	12-month period immediately following implementation of triage was observed as a transition period to the new triage system and was excluded from the final analysis. No triage unless all 4 variables available or other indication provided.	Referral if significant criteria OR KFRE 5-year risk >3%. If no criteria met then back to referrer with "low risk letter". If risk 3-10% book as non-urgent (< 6 months) If risk >10% book as urgent (< 4 weeks).	Pre-triage - No set criteria. Referrals triaged as urgent, non-urgent, or "do not book" by an individual rotating nephrologist.	Compared between pre and post triage periods: 1. Wait time between referral and nephrology visit (days). 2. Number of consults.	Pre-triage 1st Jan 2011 to 31st Dec 2011. Post-triage 1st Jan 2013 to 31st Dec 2013
Hong et al. (2020)[20 b]	Primary to nephrology care, New South Wales, Australia	Pre-post	Patients referred to St George Hospital Renal Department, n/a	KFRE for 5-year risk, unclear on number of variables	n/a	In Jan 2019 triage consultant and risk-based triage introduced. Referral if KFRE 5-year risk >3%. If < 3% risk, referral on consultant discretion.	Pre-triage periods in 2018 and 2017.	Number of consults between pre- and post- triage periods.	2017 to 2019
Smekal et al. (2019)[3 a]	Multidisciplinary to general nephrology care, Alberta, Canada	Mixed-methods	Interviews, 27: Southern Alberta Renal Program, 9 patients & 1 family member; multidisciplinary CKD healthcare providers in Calgary, Alberta, 17. Survey in Calgary, Alberta: patients from CKD clinics, 413; CKD healthcare providers, 73.	KFRE for 2-year risk, unclear on number of variables .	Interviews: Patients age >18 years with non-dialysis CKD, discharged from multidisciplinary to general nephrology care & multidisciplinary CKD healthcare providers (nephrologists, nurses, allied health professionals). Survey: All patients from CKD multidisciplinary and general nephrology clinics & all multidisciplinary CKD healthcare providers.	Transition to CKD multidisciplinary clinic when KFRE 2-year risk ≥ 10% or eGFR ≤ 15 ml/min/1.73m^2 implemented in 2017. Interviews post-implementation. Surveys post-implementation: patient – paper-based care experience survey & provider – online job satisfaction survey.	Surveys pre-implementation: patient – paper-based care experience survey & provider – online job satisfaction survey.	Interview data collection stopped once data saturation reached. Survey responses returned within time period.	Survey distributed to patients Nov 2016-Jan 2017 (pre-implementation) and Jan 2018-Mar 2018 (post implementation. Survey responses from providers Nov 2016-Jan 2017 (pre-implementation) and Apr 2018 (post-implementation).
Che et al. (2020)[19 b]	Discharges from multidisciplinary	Retrospective cohort	Prevalent CKD patients in MCKC in 2013 with	4- variable KFRE for 2-	Included CKD patients from MCKC with available data.	Revised eligibility criteria between 2016 and 2018 from eGFR < 30ml/min/1.73m^2 to	n/a	Number discharged from MCKC, re-referred,	2013 to Jan 2020

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	care, Ontario, Canada		available data, 643.	and 5-year risk		eGFR <15ml/min/1.73m^2 & KFRE 2-year risk > 10%. If not eligible then discharged from MCKC.		commenced RRT, and died.	
Sendak et al. (2016)[21 b]	Private care, North Carolina, US	Prospective cohort	Duke Connected Care a MSSP, 413.	KFRE for 2-year risk, unclear on number of variables	Patients alive and without evidence of ESKD. Excluded if had past nephrology visit and those with AKI without chronic dysfunction.	Patients with a KFRE 2-year risk >15% referred for "population health rounding" – in-depth EHR weekly review with MDT to decide on changes in management	n/a	Number of patients rounded per month, time per case during rounds, % of patient at rounds that have action taken, incidence of ESKD, RRT modality, number of dialysis crash-starts.	June 2015 for 5 months

N – number; KFRE – Kidney Failure Risk Equation; n/a – not available; CKD – Chronic kidney disease; eGFR – estimated glomerular filtration rate; MCKC – Multi-Care-Kidney-Clinic; RRT – renal replacement therapy; MSSP – Medicare Shared Savings Program; AKI – acute kidney injury; EHR – electronic health record; MDT – multidisciplinary team; ESKD – end-stage kidney disease.

[a] Full text

[b] Conference abstract

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Table 2: Results and critical appraisal of included studies

Study	Baseline characteristics: Age (years)	Baseline characteristics: Sex (male [%])	Baseline characteristics: eGFR (ml/min/1.73m^2)	Baseline characteristics: Urine ACR	Results	Key findings	Strengths	Limitations	Additional Points from Critical Appraisal
Hingwala et al. (2017)[a]	Pre-triage: not presented Post-triage: 68 (median)	Pre-triage: not presented Post-triage 50.3	Pre-triage: not presented Post-triage: 39 (median)	Pre-triage: not presented Post-triage: 9.2 (median)	Pre-triage: - Referrals booked for appointment: not presented - Median monthly referrals: 68/month (range 44-76) - Median wait time: 230 days (range: 126-355) Post-triage: - Referrals booked for appointment: 66% - Median monthly referrals: 94/month (range 61-147) - Median wait time 58 days (range: 48-69)	Monthly referral increase of 45%. Median wait times improved (P < 0.001)	Low-cost triage system Triage process allows for clinical judgment, such that 'low-risk' patients according to KFRE are identified for nephrology review where necessary	Lack of data presented for pre-triage period (e.g baseline characteristics, number of booking) to allow for detailed historical comparison (pre-triage) No clinical or economic impact assessment of implementing KFRE triage system No long-term follow-up on patients triaged as low risk. May not be applicable to teams operating under capacity without wait lists.	Washout period for 1 year when implementing criteria Additional capacity to see referrals (additional nephrologist hired in Sept 2013, physician-assistant led clinic from Sept 2012) "Hawthorne effect" 3 centres No financial support. Authors declared no competing interests.
Hong et al. (2020)[a]	Not presented	Not presented	Not presented	Not presented	Compared to 2018 when KFRE-based triage implemented: - 25% fewer consults in 2019 (30% less than in 2017) - Fewer patients had a low-risk KFRE at triage (46% vs. 48%) - Fewer low-risk patients had clinic follow-up (50% vs. 52%) More low-risk patients remaining in clinic (86% vs. 60%) had alternative reasons for follow-up (i.e. eGFR < 30 mL/min, moderate proteinuria, or uncontrolled hypertension)	Implementing the KFRE based triage system reduced overall and low-risk patient numbers in outpatient clinics.	Triage system allowed for consultant discretion for low-risk patients to be followed up	No baseline characteristics presented for patients either pre or post implementation of the KFRE based triage process Low-risk patients with eGFR < 30 or moderate proteinuria still followed up Inability to calculate KFRE score for referrals at triage (28% in 2018, 36% in 2019) due to missing urine ACR	Limited data therefore unable to compare if pre- vs post-triage groups were similar Single centre Unclear if results "significant" No comment on what was done for patients with missing ACR
Smekal et al. (2019)[b,c]	Pre-KFRE implementation: Patients: - survey 10% < 50; 28% 50-64; 22% 65-74; 39% ≥ 75. Healthcare providers: not presented. Post-KFRE implementation: Patients: - interviews 10% 50-64; 30% 65-74; 60% ≥ 75. - survey 17% < 50; 19% 50-64; 27% 65-74; 38% ≥ 75.	Pre-KFRE implementation: - Patients (survey): 60. - Healthcare providers (survey): 28. Post-KFRE implementation: Patients: - interviews 50 - survey 57. Healthcare providers - interviews 29 - survey 27.	Not presented	Not presented	Interviews: - 9/23 (39%) patients & 17/75 (23%) providers interviewed. - Five themes were identified among patients and providers and two additional categories identified among providers only. Questionnaire: - Majority of patients satisfied with their care in both periods with no overall differences. However, there were improvements in patients' experience of access to care, caring staff, and safety of care. - Of the 75 providers 40 (53%) and 33 (44%) completed the pre- and post-implementation job satisfaction survey, respectively; no differences in providers'	Patients and healthcare providers reported - improved the focus of MDT clinics by targeting high-risk patients - using KFRE to target care to high-risk patients was a key strength. Enhanced the sustainability of the clinics Providers expressed concern that there may be inadequate access to and lower quality of care for low-risk patients discharged from multidisciplinary care, although patients did not experience this.	Includes both patient and provider perspectives. Presents both qualitative and quantitative data.	More healthcare providers than patients in qualitative component; data saturation was achieved in both groups. Most patients interviewed had >5 years in MDT care prior to discharge so may not be representative of general nephrology patients. Patients discharged from MDT clinics within previous 12 months so limited time period following implementation of risk approach. Unable to pair pre- and post- survey responses or establish response rate. Single centre.	Limited to English speaking participants. Low response rate. Detailed information on data collection and analysis, questions provided. Mostly female providers. Funder had no role in the project. Authors declared no competing interests.

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	Healthcare providers: not presented.				job satisfaction.	No difference in patients' care experience. No difference in healthcare provider job satisfaction.			
Che et al. (2020)[d]	Not presented	Not presented	< 30	Not presented	470 (73%) continued follow-up in MCKC. Of 142 (22%) discharged to primary care: - 52 (37%) died - 15 (11%) re-referred to nephrology (at median 982 [IQR 560] days) - 8 (6%) initiated RRT (at median 850 [1411]) days; 5 [63%] for unforeseen acute illness) 31 (5%) discharged to general nephrology.	Discharge of a significant number of patients when moving to new criteria, few of whom ultimately required RRT that could have been prevented.	"Low loss to follow-up".	Completeness of data and follow-up unclear. May not be generalisable as based on one regional renal program.	Unclear reason for chosen threshold or if this is the ideal level. Limited data No comparison or control group. No detail on missing data
Sendak et al. (2016)[d]	Not presented	Not presented	Not presented	Not presented	Of 335 patients of 413 eligible: 73 (21.8 %) management changes: - 53 (72.6%) to nephrology - 8 (11.0%) to primary care - 7 (9.6%) lab recommendations - 4 (5.5%) medication recommendations. Average time-per-case per health round 2m 12 s. Of the remaining 262 (78.2%) patient screening did, however, identify; - 110 (42.0%) seeing a nephrologist - 35 (13.3%) recently deceased - 25 (9.5%) on dialysis.	Patients with CKD at high risk of progression to ESKD can be identified using validated algorithms applied to structured data that is readily available. A significant proportion of patients identified in this way require management changes, including patients that require nephrology review. Limited ability to screen out patients using structured data. Health rounds can be performed relatively rapidly.	KFRE easily applicable Highlights additional patients for review in primary/speciality care.	No data on long term outcomes. Data from one renal program, therefore, may not be generalisable.	Elderly patients Later months more efficient as workflow optimised – not taken into consideration for time per case. Private healthcare setting so not generalisable.

eGFR – estimated glomerular filtration rate; ACR – albumin:creatinine ratio; KFRE – Kidney Failure Risk Equation; MDT – Multidisciplinary team; MCKC – Multi-Care-Kidney-Clinic; RRT – renal replacement therapy; CKD – chronic kidney disease; ESKD – end-stage kidney disease.

Appraisal tool used for the studies:

[a] Joanna Briggs Institute Quasi-experimental study checklist

[b] CASP Qualitative Studies Checklist (interviews)

[c] Centre for evidence-based management critical appraisal of a survey

[d] CASP Cohort Study Checklist

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Primary care to speciality care interface

Two studies[18,20] reported outcomes following the introduction of a new referral process incorporating a 5-year ESKD risk threshold of >3% to triage patients from primary care to speciality care; although they also incorporated additional referral criteria for eligibility for speciality care review. Both studies measured the number of consults pre- and post-implementation of the KFRE: Hingwala et al. found monthly referrals increased by 45%[18], whereas Hong et al. found referrals decreased by 25-30%.[20]

Hingwala et al. also showed a significant reduction in the wait-time from referral to review (median 58 vs 230 days) following implementation of the new triage system.[18]

General nephrology to multidisciplinary/advanced kidney care clinic

Patients under nephrology care may be managed in a general nephrology clinic or, for patients with more advanced CKD, a multidisciplinary clinic focused on managing CKD complications and RRT preparation. Two studies examined the use of the KFRE at this interface and introduced similar eligibility criteria for entry into and management in the multidisciplinary clinic: a 2-year ESKD risk > 10% or eGFR < 15 ml/min/1.73m^2 (formula used for calculating eGFR not specified in the studies).

Che et al.[19] found that by applying these criteria to all prevalent patients in the multidisciplinary clinic, 73% remained in the clinic, 5% were stepped down to general nephrology care, and 22% were discharged to primary care. Of the latter, 11% required re-referral to nephrology, and 6% ultimately required RRT, although the majority in the context of an unforeseen acute illness.

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Smekal et al.[3] evaluated patient and healthcare providers' views and experiences following the implementation of the new KFRE-based criteria for multidisciplinary clinic management. Providers felt using KFRE to target high-risk patients was a key strength, limiting inappropriate referrals and improving the clinic's focus. The caseload was felt to be more 'acute', but overall workload not significantly changed, and there was no significant difference in job satisfaction. Providers also expressed concern that there may be inadequate access to and a lower quality of care for low-risk patients discharged from the multidisciplinary clinic, such as with education and monitoring. Some nurses reported discharged patients were contacting them for test results, and patients reported missing this aspect of their care. Although there was some improvement in patient's experience of access to care, caring staff, and safety of care, most patients were satisfied with their care both pre- and post-implementation, and there was no difference in patients' overall care experience.

Treatment planning

The study by Sendak et al.[21] took place in an "Accountable Care Organisation" within the Medicare Shared Savings Program (Medicare is a health insurance program mainly for people aged ≥65 years in the US). A 2-year ESKD risk threshold of >15% was applied to the cohort to identify high-risk CKD patients, not already under nephrology review, for a multidisciplinary review of their electronic healthcare record (EHR). The reviews were able to be performed relatively rapidly (average 2 minutes 12 seconds) and led to changes in management in 21.8% of cases, most often a referral for nephrology review.

Critical appraisal

All studies were limited to a single centre or region. No randomised control trials were identified. Two were cohort studies[19,21], one was a mixed-method study[3] and two were quasi-

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experiment studies[18,20]. All studies stated a clear aim or issue to be addressed. Sample size varied depending on the study method, and two studies did not state it.[18,20] Participant baseline characteristics data was not available in two studies[19,20] and lacking in one study[18] where pre-intervention information was not available. For the qualitative aspect of the mixed-methods study[3], the patients selected were English-speaking only. Although this may have allowed for simpler data analysis as there was no requirement for language translation, this selection bias may limit the perspectives obtained of those experiencing the intervention. In this same study, the survey response rate for patients was unclear, and for providers, no data was provided to establish any difference in demographics to non-responders. Although mostly proportions were reported in the data analysis of quantitative studies, there was a lack of reporting of statistical significance, strength of any associations or measures of precision in most. Three studies[19-21] reported only proportions when applying the intervention. These three studies had not commented on the number of patients, if any, who could not be assessed for risk due to missing data or whether there were any confounding factors.

DISCUSSION

In this systematic review, we identified five studies of varying methodologies evaluating the actual impact of utilising the KFRE in clinical practice for patients with CKD. The equation was utilised in three main areas: i) triaging patients between primary and speciality nephrology care, ii) directing patients between general nephrology and multidisciplinary advanced CKD care clinics, and iii) treatment planning where high-risk patients within a healthcare program were identified for a multidisciplinary EHR review.

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Interpretation of the results in the context of other evidence

The findings of previous studies exploring the potential impact of applying the KFRE at the primary care-speciality care interface suggest referral numbers could potentialy increase depending on the threshold criteria set.[5,22,23] This is consistent with the findings of Hingwala et al.[18], who reported an increase in referrals. However, wait times improved, possibly related to additional capacity to see referrals and a "Hawthorne effect" (better performance as a result of health care professionals' awareness of being observed in a study).[24] In contrast, Hong et al.[20], who implemented the same risk threshold, reported a fall in referrals. These contrasting findings likely reflect differences in local practice and case-mix, whereby a higher proportion of low-risk patients (i.e. older, higher eGFRs, lower urine ACRs) were managed by nephrology rather than by primary care before implementation of the KFRE criteria, compared to the centre reported by Hingwala et al. We were unable to review any population differences pre- and post- triage or between studies due to a lack of reported data.[18,20]

In specialist nephrology care, many centres have advanced kidney care clinics for those patients who require increased multidisciplinary support to manage CKD complications and prepare for RRT.[25,26] The KFRE has the potential to identify patients at higher risk of ESKD to more efficiently direct increased support and resources for this group. Those identified as low risk can be stepped down, reducing clinic burden and unnecessary interventions. Two studies evaluated this part of the CKD pathway and applied similar eligibility criteria for multidisciplinary clinic management. Che et al.[19] found many patients could be discharged from the multidisciplinary clinic, the majority back to primary care, with only a small number referred back to nephrology care or ultimately requiring RRT.

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The findings of Smekal et al.[3] suggest some anxiety regarding reduced access to services, education and monitoring when patients are discharged from the multidisciplinary clinic to general nephrology care based on their KFRE-calculated risk. Maintaining patient and provider satisfaction with CKD care when using the KFRE to discharge patients is an important issue that will require the considered configuration of local renal service delivery to ensure accessibility and patient safety. It remains unknown if the application of the KFRE to identify higher-risk patients for multidisciplinary care impacts key outcomes such as progression of CKD, commencing RRT, cardiovascular events or mortality.

Sendak et al.[21] reviewed the impact of utilising KFRE in a private healthcare setting in a primarily elderly population to identify patients who required input from medical teams (in primary or specialist care), treatment changes or additional lab testing. The results of this study may not be easily generalisable as this is not a widely used provider model in some countries.

Other suggested areas of use have been for planning RRT in an elderly population where competing risk of death is of concern[27] and for creation of vascular access[28] but no studies have implemented and evaluated this.

Limitations of the evidence included in the review and the review process

The studies varied widely in methodologies used. The benefit of this mixed-methods review is a greater breadth of understanding and evidence around the application of the risk-based KFRE in clinical practice. This can provide a more rounded body of evidence to inform changes in clinical practice and policy decision making. No randomised control trials were identified.

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Study quality also ranged widely. Some studies did not provide adequate detail of their population characteristics. As a result, it was difficult to assess if the population was representative or findings could be generalised. Recognising differences between study populations was also limited by the lack of data. Few studies adequately completed statistical analysis and so it was challenging to decipher significant findings. Authors of the identified studies were not approached for additional information and we recognise this as a limitation.

Implication of the results for practice, policy, and future research

Whilst this study has identified the ease of use of the KFRE to triage patients and highlight those who would benefit from changes in management, plus the potential effect on the number of referrals and patients' and providers' experience and perspectives, the impact on health outcomes and economic benefit are still unknown.

More high-quality studies are needed to confidently support the widespread use of KFRE, particularly randomised-control trials with a focus on health outcomes and economic impact. Four trials that are in currently progress will assess the outcomes of applying the KFRE (Table 3). The Kidney CHAMP study[29], a cluster randomised controlled trial in Pittsburgh, US, will review the effectiveness of integrating the KFRE into EHR to identify patients at high risk of progression who require intervention such as nephrology guidance, medication reviews and targeted CKD education. Outcomes assessed will be a composite of a 40% reduction in eGFR and ESKD. A multicentre cluster randomised control trial in Canada[30] aims to review the risk-based approach in the primary care setting. The intervention, providing patients and providers in primary care with a patient's KFRE score and the risk-based referral criteria, will be reviewed to assess if it improves appropriate patient

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management, improves a patient's CKD-specific health literacy and affects the cost of care compared with usual care. The PREPARE NOW study[31], a cluster randomised control trial within a nephrology speciality care setting in Pennsylvania, US, is applying the KFRE amongst a suite of digital tools integrated with EHRs to alert the health care team of patients at risk of progression and need for intervention. This will be in addition to multiple other components, and patient-reported, biomedical and health system outcomes will be collected. Early findings support the "ease of use" and "helpfulness" of the tools.[32] The fourth trial, with details described in the protocol by Hemmelgarn et al.[33], is a multiphase mixed-methods study. Following the completion of phase 2, findings have been published by Smekal et al.[3], a study which met the criteria for inclusion in this systematic review. The whole trial took on a pre-post design and applied the KFRE to patients in nephrology multidisciplinary CKD clinics in Alberta, Canada. The final phase will aim to evaluate the costs of care and outcomes before and after the introduction of risk-based triage, such as health care resource use, frequency of testing, modality choice, and death.

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Table 3. Randomised-control trials currently in progress

Study	Healthcare setting, Country	Study design	Population, N	KFRE details	Inclusion/exclusion criteria	Intervention	Control	End-points	Follow-up
Jhamb et al. (2019) - The Kidney CHAMP Study[29]	Primary care, Pittsburgh, US	Cluster RCT	1650 high-risk CKD patients	Not presented	Patients with high-risk CKD (as defined by validated risk prediction models or by current eGFR value or recent decline in eGFR values) Inclusion: Age 18 to 85 years, eGFR <60ml/min/1.73m^2 Exclusion: History of renal transplant, on maintenance dialysis, recent (within 12 months) outpatient nephrology visit, baseline eGFR less than 15ml/min/1.73m^2, expected survival < 6 months, active substance dependence or severe/uncontrolled psychiatric condition	EHR-based PHM intervention: Nephrologist-led E-consults, pharmacist-led medication reviews, and nurse-led CKD education.	Usual care	Primary outcome is a composite of 40% reduction in eGFR or ESKD Secondary outcomes: improved hypertension control, use of RAASi, and avoidance of renally contraindicated medications	42 months
Harasemiw et al. (2019)[30]	32 Primary care clinics, Manitoba and Alberta, Canada	Multicentre cluster RCT	Estimate each clinic to have 185 patients with CKD	Not presented	Inclusion: Aged 18 years and older wit CKD G3-G5 attending the participating clinics	Active knowledge translation intervention: addition of KFRE and decision aids to clinics' Data Presentation Tool, patient-facing visual aids, a medical detailing visit, sentinel feedback reports.	Usual care: exposed to current guidelines for CKD management, without active dissemination.	Primary outcomes: proportion of patients with measured urine ACR, and proportion of patients appropriately treated with ACEi or ARB. Secondary outcomes: the optimal management of diabetes, hypertension and cardiovascular risk; prescriptions of NSAIDs; and decline in eGFR. Substudy: measure patients' CKD-specific health literacy and trust in physician care via surveys administered in the clinic post-visit; measure provider satisfaction with the risk prediction tools; at the health system level, outcomes including cost of CKD care, and appropriate referrals for patients at high risk of kidney failure.	Primary and secondary outcomes reviewed at 1 year after the intervention implementation. Exception for decline in eGFR, which will be measured 2 years post-intervention.
Green et al. (2018) - PREPARE NOW Study[31]	Nephrology care, Geisinger Health System kidney specialty clinics, Pennsylvania, US	Cluster RCT	1572 participants	8-variable KFRE 2-year risk score	Inclusion: Patients currently receiving care at Geisinger nephrology practices, aged 18 and older with advanced kidney disease determined by eGFR or presence of albuminuria.	Implement new electronic health information tools (disease registry and risk prediction tools) to help providers recognise patents in need of Kidney	Usual care	Primary outcomes: change in % patients feeling in control of their decision-making, change in number of hospitalisations, change in % patients with advance directives for kidney care.	36 months

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						Transitions Care. Implement a Kidney Transitions Specialist who will provide and facilitate integrated delivery of patient support programs.		Secondary outcomes: change in % self-care patients with biomedical care plans, change in % patients with values aligned care, change in % patients with preferences for renal replacement therapy documented, change in % patients with emergency dialysis initiation, change in months to kidney failure, change in % patients with vascular access (e.g., fistula) in place at dialysis initiation.	
Hemmelgarn et al. (2018)[33]	Nephrology multidisciplinary CKD clinics, Alberta, Canada	Pre/post cohort	Not presented	Not presented	Inclusion: Adults aged 18 years and older with sustained eGFR <30 mL/min/1.73 m^2, who are followed by a nephrologist. Exclusion: Patients receiving dialysis or with a kidney transplant prior to the study period.	Transition to CKD multidisciplinary clinic when KFRE 2-year risk ≥ 10% or eGFR ≤ 15 ml/min/1.73m^2.	Pre-triage period	Clinical outcomes (hospitalisation and emergency department visits, death), use of modalities that improve patient experience and outcomes (home dialysis, kidney transplantation), resource use (physician visits, laboratory tests), process-based quality indicators for appropriate CKD care (assessment of albuminuria, use of ACE-I/ARBs in those with albuminuria, and statins), costs and proportion of patients risk-stratified and appropriately managed.	Cohort accrual for the pre-period from April 2015 to April 2016. Post-period, cohort accrual from April 2017 to April 2018, with follow-up to April 2019.

KFRE – Kidney Failure Risk Equation; RCT – Randomised control trial; CKD – chronic kidney disease; eGFR – estimated glomerular filtration rate; EHR – electronic health record; PHM – Population Health Management; ESKD – end-stage kidney disease; RAASi – renin angiotensin aldosterone system inhibitors; ACEi – Angiotensin-Converting Enzyme inhibitor; ARB - Angiotensin Receptor Blocker; NSAID - Non-Steroidal Anti-inflammatory Drugs.

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Despite the lack of sufficient impact studies, strong evidence to date from validation studies and studies investigating potential impact in clinical practice have been encouraging. As a result, several CKD guidelines have or are in the process of incorporating KFRE risk-based criteria in their pathways.[6,7,34]

Conclusion

The KFRE has been extensively validated, but there has been relatively little evaluation of its clinical and economic impact. This is the first study to systematically review studies exploring at the actual impact of using the KFRE in clinical practice. Additional high-quality studies are required, and trials assessing the impact of using KFRE at various stages across the CKD pathway are in progress.

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The views expressed are those of the author(s) and not necessarily those of the NIHR or the Department of Health and Social Care.

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AUTHOR CONTRIBUTIONS

HKB, PC, MC and DK conceived the study. HKB developed the methodology with guidance from PC, MC, DK, OLA and AF. The database, reference and citation searches were completed by HKB and AF. Screening and selection of the included studies was conducted by HKB and AF; PC as the third reviewer was consulted for any disagreements. HKB and AF critically appraised the included studies. HKB drafted the initial manuscript and all authors contributed to revision and approval of the final manuscript.

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ETHICAL APPROVAL STATEMENT

Ethical approval not required for this systematic review.

REFERENCES

1. Bikbov B, Purcell CA, Levey AS, et al. Global, regional, and national burden of chronic kidney disease, 1990–2017: a systematic analysis for the Global Burden of Disease Study 2017. *The Lancet.* 2020;395(10225):709-733.
2. Daker-White G, Rogers A, Kennedy A, Blakeman T, Blickem C, Chew-Graham C. Non-disclosure of chronic kidney disease in primary care and the limits of instrumental rationality in chronic illness self-management. *Social science & medicine (1982).* 2015;131:31-39.
3. Smekal MD, Tam-Tham H, Finlay J, et al. Patient and provider experience and perspectives of a risk-based approach to multidisciplinary chronic kidney disease care: A mixed methods study. *BMC Nephrology.* 2019;20(1).
4. Chronic kidney disease in adults: assessment and management (CG182). In. London: National Institute for Health and Care Excellence (UK); 2014.
5. Bhachu H, Cockwell P, Subramanian A, et al. Impact of utilising risk-based stratification on referral of patients with chronic kidney disease from primary care to specialist care in the UK. *Kidney International Reports.* 2021;6(8):2189-2199.
6. Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group: KDIGO 2012 Clinical Practice Guideline for the Evaluation and Management of Chronic Kidney Disease. Kidney Inter suppl, 3: 1-150, 2013.

For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

7. Farrington K, Covic A, Aucella F, et al. Clinical Practice Guideline on management of older patients with chronic kidney disease stage 3b or higher (eGFR <45 mL/min/1.73 m^2). *Nephrology Dialysis Transplantation.* 2016;31(suppl_2):ii1-ii66.

8. Ramspek CL, de Jong Y, Dekker FW, van Diepen M. Towards the best kidney failure prediction tool: a systematic review and selection aid. *Nephrol Dial Transplant.* 2020;35(9):1527-1538.

9. Tangri N, Grams ME, Levey AS, et al. Multinational Assessment of Accuracy of Equations for Predicting Risk of Kidney Failure: A Meta-analysis. *JAMA.* 2016;315(2):164-174.

10. Ramspek C, De Jong Y, Dekker F, Van Diepen M. Predictive models for progression of chronic kidney disease to kidney failure: A systematic review. *Nephrology Dialysis Transplantation.* 2018;33 (Supplement 1):i155.

11. The Kidney Failure Risk Equation. Accessed 30/10/2017.

12. Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate health care interventions: explanation and elaboration. *PLoS Med.* 2009;6(7):e1000100.

13. Tangri N, Stevens LA, Griffith J, et al. A predictive model for progression of chronic kidney disease to kidney failure. *JAMA.* 2011;305(15):1553-1559.

14. Critical Appraisal Skills Programme. https://casp-uk.net/. Accessed 14/4/2021.

15. Joanna Briggs Institute Critical Appraisal Tools. https://jbi.global/critical-appraisal-tools. Accessed 14/4/2021.

16. Critical Appraisal of a Survey. Center for Evidence-Based Management (CEBMa). https://www.cebma.org/wp-content/uploads/Critical-Appraisal-Questions-for-a-Survey.pdf. Accessed 14/4/2021.

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17. Popay J, Roberts H, Sowden A, et al. Guidance on the Conduct of Narrative Synthesis in Systematic Reviews: A Product from the ESRC Methods Programme. 2006.

18. Hingwala J, Wojciechowski P, Hiebert B, et al. Risk-Based Triage for Nephrology Referrals Using the Kidney Failure Risk Equation. *Can J Kidney Health Dis.* 2017;4:2054358117722782.

19. Che M, Thanabalasingam SJ, Iliescu EA, White CA. Patient outcomes following discharge from a CKD clinic. *Journal of the American Society of Nephrology.* 2020;31:227.

20. Hong R, Pirabhahar S, Turner K, Katz IJ. Triage system for nephrology referrals using the kidney failure risk equation (KFRE) score. *Nephrology.* 2020;25 (SUPPL 3):53.

21. Sendak M, Cameron B, Komives E, et al. Developing a data-driven work flow for population health rounding. *Journal of General Internal Medicine.* 2016;31(2):S874-S875.

22. Slattery LM. Applying novel kidney failure risk equations applications to new patient referrals: A 6-month retrospective review. *Journal of the American Society of Nephrology.* 2019;30:1173.

23. Major RW, Shepherd D, Medcalf JF, Xu G, Gray LJ, Brunskill NJ. The kidney failure risk equation for prediction of end stage renal disease in UK primary care: An external validation and clinical impact projection cohort study. *PLoS Medicine.* 2019;16(11).

24. McCambridge J, Witton J, Elbourne DR. Systematic review of the Hawthorne effect: new concepts are needed to study research participation effects. *J Clin Epidemiol.* 2014;67(3):267-277.

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25. Nicoll R, Robertson L, Gemmell E, Sharma P, Black C, Marks A. Models of care for chronic kidney disease: A systematic review. *Nephrology (Carlton).* 2018;23(5):389-396.

26. Evans M, Lopau K. The transition clinic in chronic kidney disease care. *Nephrol Dial Transplant.* 2020;35(Suppl 2):ii4-ii10.

27. Nastasa A, Apetrii M, Onofriescu M, et al. Risk prediction for death and end-stage renal disease does not parallel real-life trajectory of older patients with advanced chronic kidney disease-a Romanian center experience. *Nephrology Dialysis Transplantation.* 2020;35 (SUPPL 3):iii486.

28. Inston N, Lok CE. Improving precision in prediction: Using kidney failure risk equations as a potential adjunct to vascular access planning. *Journal of Vascular Access.* 2019;20(1):95-97.

29. Jhamb M, Yabes J, Fischer G, et al. Electronic health record based population health management for improving CKD care: the kidney-coordinated health management partnership (CHAMP) study. *Journal of the American Society of Nephrology : JASN.* 2019;30:525-.

30. Harasemiw O, Drummond N, Singer A, et al. Integrating Risk-Based Care for Patients With Chronic Kidney Disease in the Community: Study Protocol for a Cluster Randomized Trial. *Canadian Journal of Kidney Health and Disease.* 2019;6(no pagination).

31. Green JA, Ephraim PL, Hill-Briggs FF, et al. Putting patients at the center of kidney care transitions: PREPARE NOW, a cluster randomized controlled trial. *Contemporary Clinical Trials.* 2018;73:98-110.

For peer review only

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32. Green JA, Ephraim PL, Hill-Briggs F, et al. Integrated Digital Health System Tools to Support Decision Making and Treatment Preparation in CKD: The. *Kidney Medicine.* 2021;3(4):565-+.

33. Hemmelgarn BR, Smekal MD, Weaver RG, et al. Implementation and evaluation of a risk-based approach to guide chronic kidney disease care: Protocol for a multiphase mixed-methods study. *Canadian Journal of Kidney Health and Disease.* 2018;5(no pagination).

34. Chronic kidney disease: assessment and management (NG203). London, National Institute for Health and Care Excellence (UK); 2021.

FIGURES

Figure 1: Flow diagram to show the studies identified from searches

(See separate file – "FIGURE 1")

SUPPLEMENTARY MATERIAL TABLE OF CONTENTS

Table S1: Search strategies for the searches

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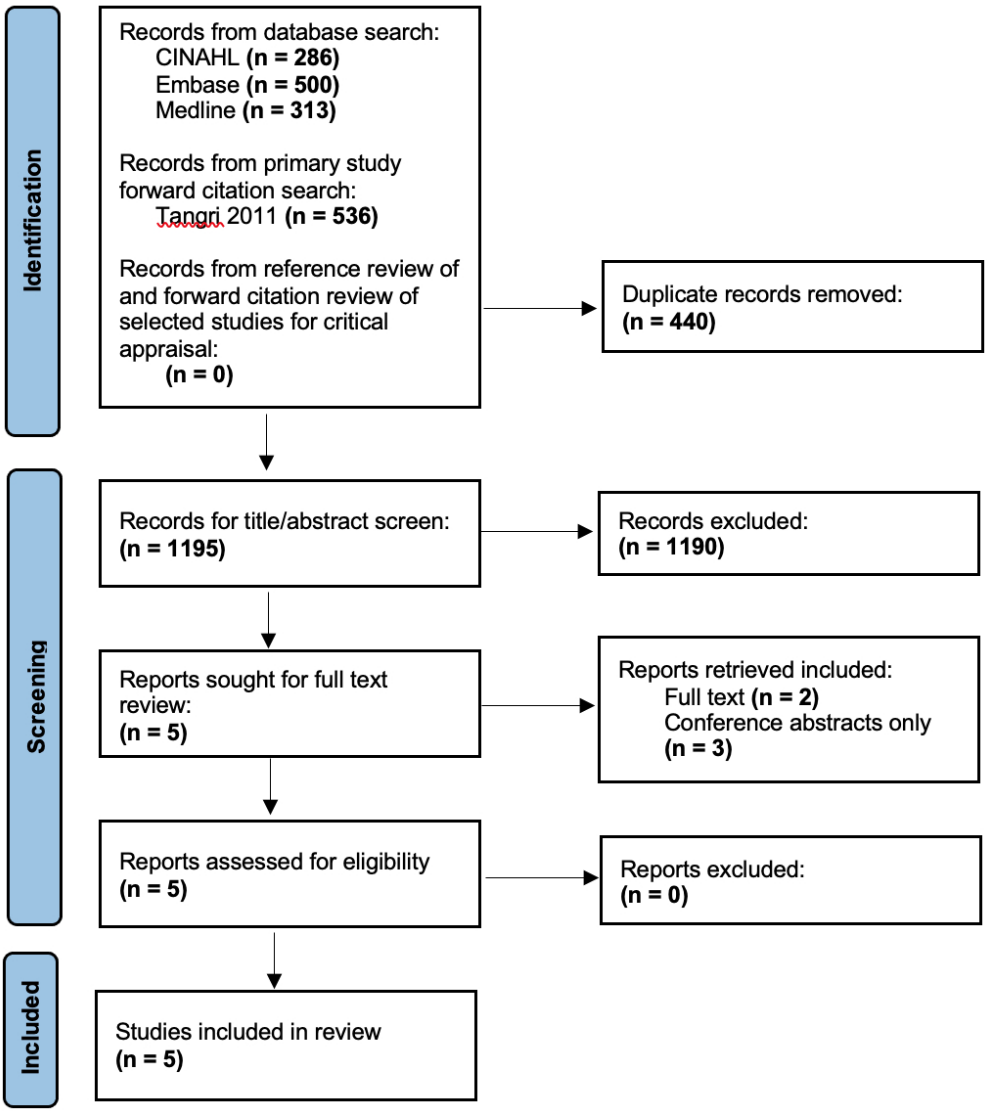


Figure 1: Flow diagram to show the studies identified from searches

180x202mm (144 x 144 DPI)

SUPPLEMENTARY MATERIAL

Table S1: Search strategies for the searches

Database	Search strategy/ search terms	Results
MEDLINE (Ovid) ALL* 1946 to October 15, 2021	1. (CKD or CRF or CRD or CKF).ti,ab.	53092
	2. (renal insufficiency or kidney failure or kidney disease* or renal disease* or renal failure).ti,ab.	242663
	3. exp Renal insufficiency, Chronic/	122966
	4. 1 or 2 or 3	304342
	5. (kidney failure risk* or KFRE).ti,ab.	95
	6. 4 and 5	95
	7. limit 6 to yr="2011 -Current"	90
	8. (kidney failure adj3 risk*).ti,ab.	372
	9. KFRE.ti,ab.	34
	10. 8 or 9	372
	11. 4 and 10	372
	12. Limit 11 to yr="2011 -Current"	313
Embase (Ovid)* 1974 to 2021 October 15	1. (CKD or CRF or CRD or CKF).ti,ab. [84422]	90073
	2. (renal insufficiency or kidney failure or kidney disease* or renal disease* or renal failure).ti,ab.	358143
	3. exp chronic kidney failure/	116396
	4. 1 or 2 or 3	423454
	5. (kidney failure risk* or KFRE).ti,ab.	164
	6. 4 and 5	164
	7. limit 6 to yr="2011 -Current"	159

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	8. (kidney failure adj3 risk*).ti,ab.	566
	9. KFRE.ti,ab.	71
	10. 8 or 9	567
	11. 4 and 10	567
	12. Limit 11 to yr="2011 -Current"	500
CINAHL Plus* (EBSCOhost)	1. TI (CKD OR CKF OR CRD OR CRF) OR AB (CKD OR CKF OR CRD OR CRF) [11577]	12575
	2. TI (renal insufficiency OR kidney failure OR kidney disease* OR renal disease* OR renal failure) OR AB (renal insufficiency OR kidney failure OR kidney disease* OR renal disease* OR renal failure) [51205]	54357
	3. (MH "Renal Insufficiency, Chronic+") [28518]	29825
	4. S1 OR S2 OR S3 [65333]	68993
	5. TI KFRE OR AB KFRE [5]	9
	6. TI (kidney failure risk*) OR AB (kidney failure risk*) [335]	360
	7. S5 OR S6 [335]	360
	8. S4 AND S7 [335]	360
	9. S4 AND S7 Limiters - Publication Year: 2011-2021[260]	286
Web of Science*	Tangri 2011 A predictive model for progression of chronic kidney disease to kidney failure	536
Forward citation review	Selected studies	0
Reference review	Selected studies	0

*Search date 18/10/2021

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Prisma Checklist

Section and Topic	Item #	Checklist item	Location where item is reported
TITLE			
Title	1	Identify the report as a systematic review.	Page 1
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	Pages 2-3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	Page 5
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	Page 5
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	Pages 6-7
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	Page 6, Table S1
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	Table S1
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	Pages 6-7
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	Page 7
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	Page 7
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear information.	Page 7
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	Page 8
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Page 9
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	Page 8
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	NA
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Page 8

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Section and Topic	Item #	Checklist item	Location where item is reported
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	NA
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Page 8
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	NA
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	Page 8
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Page 8
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	Page 8, Figure 1
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	Figure 1
Study characteristics	17	Cite each included study and present its characteristics.	Pages 8-10, Table 1
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	Table 2
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	Table 2
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	Table 2
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Table 2
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Page 16-17
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	NA
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	Page 16-17
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Page 16-17
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	Pages 18-19
	23b	Discuss any limitations of the evidence included in the review.	Pages 19-20

For Peer Review Only

Section and Topic	Item #	Checklist item	Location where item is reported
	23c	Discuss any limitations of the review processes used.	Pages 4; 19-20
	23d	Discuss implications of the results for practice, policy, and future research.	Pages 20-21
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	Pages 3 & 6
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	Pages 3 & 6
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	NA
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Page 25
Competing interests	26	Declare any competing interests of review authors.	Pages 24-25
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	NA, Data in Tables 1&2

From: Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ 2021;372:n71. doi: 10.1136/bmj.n71

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